



**Politecnico
di Torino**

POLITECNICO DI TORINO

Bachelor Degree in Computer Engineering

**Development of PMU Data Analysis Methods for
Synchrophasor Performance Evaluation and
Sub-Oscillation Detection**

Internship Technical Report

Project: 24DIT05 GridData

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Abstract

This internship technical report presents the work carried out at INRiM within the Smart Grid Data project. The activity focused on synchronized phasor measurement data, PMU and micro-PMU performance evaluation, and the development of MATLAB-based analysis methods for the detection of sub-oscillation phenomena. The work started with the study of PMU measurement principles, synchrophasor quantities, Total Vector Error, ROCOF, off-nominal conditions, harmonic distortion, amplitude modulation, and uncertainty evaluation. It then evolved into the design and testing of algorithms for detecting oscillatory regions and estimating dominant oscillation frequencies from PMU output channels.

The main contribution of the internship was the development of a MATLAB analysis workflow based on a literature review of power-system oscillation detection methods. The implemented algorithm applies a sliding-window approach, steady-state recognition, Nyquist-frequency handling, FFT-based peak detection, adaptive thresholding based on the median and MAD of the spectrum, and sinusoidal least-squares validation. The method was applied to frequency, amplitude, and phase channels and was adapted for both commercial micro-PMU data and INRiM QPMU data. The results were compared with external reference approaches, including the NPL algorithm, and with measurements collected at different reporting rates.

The internship produced Excel-based measurement evaluations, MATLAB scripts, comparison tables, and graphical outputs. The overall work contributed to a reproducible analysis workflow for Smart Grid Data studies and highlighted the importance of reporting rate, channel selection, measurement uncertainty, and robust signal-processing choices in PMU-based oscillation analysis.

1. Introduction

Modern electrical grids require accurate and synchronized measurements in order to monitor stability, detect disturbances, and support reliable operation. Phasor Measurement Units are used to provide time-synchronized estimates of voltage and current phasors, frequency, and rate of change of frequency. These measurements are especially relevant in smart grid applications because they allow dynamic events to be observed and compared across different points of the network.

The increasing complexity of power systems makes this type of measurement activity more important. Traditional power systems were mainly based on large synchronous generators and relatively predictable operating conditions. Modern grids, however, include distributed generation, renewable energy sources, inverter-based resources, power electronics, and bidirectional power flows. These elements improve flexibility and sustainability, but they also introduce new dynamic behaviours. For this reason, measurement systems must be able to capture not only steady-state quantities but also dynamic phenomena such as oscillations, modulation effects, harmonic distortion, frequency deviations, and transient changes.

The internship was carried out at INRiM, the Italian National Institute of Metrological Research, within the Smart Grid Data project. The work involved the analysis of synchronized measurement data obtained from a commercial micro-PMU and from an INRiM reference QPMU setup. The analysed data included steady-state tests, off-nominal frequency tests, amplitude modulation tests, harmonic distortion tests, and sub-oscillation test conditions. The work was connected to the broader objective of improving the evaluation and interpretation of synchronized measurement data for modern distribution grids.

The main objective was to support PMU performance evaluation and to develop MATLAB-based methods for detecting sub-oscillation patterns from PMU output quantities. The work was therefore both experimental and

computational. It included measurement analysis, literature review, algorithm design, implementation, testing, comparison, and report preparation. In particular, the algorithmic part of the internship was developed after studying scientific papers and internal project material related to oscillation detection and PMU-based signal analysis.

A relevant aspect of the project was that the analysis was not performed on raw high-frequency waveforms only. Instead, the focus was mainly on PMU output channels, such as voltage magnitude, voltage angle, frequency, and ROCOF. These quantities are already processed by the PMU and are available at a limited reporting rate, such as 10, 25, 50, or 100 frames per second. This creates practical constraints for signal analysis. For example, a PMU reporting at 50 fps cannot directly represent oscillatory components above 25 Hz without aliasing. This limitation strongly influenced the design of the developed algorithm.

2. Internship Context and Objectives

The internship took place between 17 February 2026 and 30 April 2026, for a total of 300 hours. The host institution was INRiM, and the activity was supervised by Dr. Bruno Trinchera. The academic tutor was Prof. Valentino Peluso from Politecnico di Torino. The internship was developed within the Smart Grid Data context, with a focus on synchronized measurement systems, PMU data analysis, and signal-processing methods for the detection of oscillatory phenomena.

The first phase of the internship was mainly dedicated to understanding the metrological and electrical background of the project. This included the study of updated power-system stability classifications, measurement instrument interconnection, synchronization issues, and the role of accurate data exchange in PMU-based monitoring. A particular effort was made to understand how measurement errors affect phasor estimation and how these errors are represented through metrics such as TVE and ROCOF.

The second phase focused on laboratory measurements and data elaboration. Synchrophasor measurement data were acquired from power-quality

instruments and analysed under different test conditions. Excel files were used to calculate and organize TVE values, ROCOF behaviour, uncertainty contributions, and comparison tables. The use of Excel was important in the initial phase because it allowed the intermediate steps of the calculations to be inspected and validated.

The third phase focused on MATLAB implementation. MATLAB was used both to support waveform generation and to develop the algorithms for analysing PMU outputs. The developed scripts were progressively improved to support different data formats, different PMU channels, automatic delimiter detection, robust numeric conversion, phase unwrapping, and sliding-window signal analysis. This phase represented the main software-development component of the internship.

- study PMU and micro-PMU measurement principles;
- evaluate synchrophasor performance using TVE and ROCOF;
- analyse measurement errors under steady-state and dynamic conditions;
- process experimental data using Excel and MATLAB;
- review scientific literature on oscillation detection methods;
- develop MATLAB algorithms for disturbance and sub-oscillation detection;
- extract dominant oscillation frequencies and SSO amplitudes from PMU channels;
- compare the developed INRIM algorithm with the NPL algorithm;
- compare commercial micro-PMU and INRIM QPMU results under different test conditions;
- produce a structured technical report following academic and technical-writing requirements.

From a learning perspective, the internship connected computer engineering skills with measurement science and power-system monitoring. The activity required programming, data processing, signal analysis, documentation, and interpretation of experimental results. It also required understanding the difference between implementing a working algorithm and validating whether its output is meaningful for a metrological application.

3. Technical Background

3.1 Phasor Measurement Units

PMUs are synchronized measurement devices used in modern power systems to provide real-time estimates of voltage and current phasors, frequency, and ROCOF. Time synchronization is usually obtained through GPS or GNSS systems, allowing measurements collected at different locations to be aligned to a common time reference. This synchronization is essential because a phasor measurement is meaningful only if its time reference is known accurately.

A typical PMU measurement chain includes signal acquisition, analog filtering, analog-to-digital conversion, phasor estimation, time synchronization, and data transmission. Each stage may introduce intrinsic errors. These errors can be related to the measurement chain, timing, algorithmic choices, synchronization system, quantization, hardware behaviour, or environmental conditions. For example, timing errors may affect the measured phase angle, while analog front-end errors may affect magnitude accuracy. Algorithmic errors may appear when the signal frequency deviates from nominal or when the signal contains dynamic components.

In the context of this internship, PMUs were studied not only as measurement devices but also as data sources for signal-processing algorithms. The algorithm does not directly receive the raw high-frequency waveform in most cases. Instead, it receives processed PMU outputs at a reporting rate. This distinction is important because the PMU has already performed filtering and phasor estimation before the data are analysed by the developed MATLAB scripts.

3.2 Synchrophasor Quantities

The main synchrophasor quantities considered during the internship were voltage magnitude, phase angle, frequency, and ROCOF. Voltage magnitude describes the amplitude of the phasor, while phase angle describes its angular position with respect to the time reference. Frequency describes the estimated

fundamental frequency of the signal, and ROCOF describes how quickly that frequency is changing.

In steady-state conditions, these quantities are expected to be stable. However, during dynamic events, the values may vary significantly. For example, amplitude modulation can appear in the voltage magnitude channel, while phase modulation or oscillatory behaviour may appear in the angle channel. Frequency deviations and low-frequency oscillations may appear in the frequency channel. Depending on the test condition, one channel may show the phenomenon more clearly than the others.

This channel dependence was important in the internship. The developed algorithm allows the user to select the frequency, amplitude, or phase channel. This design choice was made because no single PMU output channel is always the best for every event. In some cases, the oscillation may be more visible in the amplitude channel. In other cases, the phase channel may provide a clearer signature. The frequency channel may be useful for low-frequency dynamic phenomena but may be less informative for certain injected modulation tests.

3.3 TVE and ROCOF

Total Vector Error is a key metric for evaluating the accuracy of synchrophasor measurements. It compares the measured phasor with a reference phasor and combines magnitude and phase deviations into one percentage value. TVE was used during the internship to evaluate the behaviour of the commercial micro-PMU and the INRIM reference QPMU under different test conditions.

TVE is useful because it gives a compact measure of phasor error. A PMU may have a small magnitude error but a larger phase error, or the opposite. TVE combines both effects. For this reason, it was used in several Excel elaboration files. These files helped to compare steady-state, off-nominal, amplitude modulation, off-nominal plus amplitude modulation, and harmonic distortion conditions.

ROCOF describes the rate of change of the measured frequency. It is especially important during dynamic events, because it indicates how quickly the frequency is varying. In the internship, ROCOF and TVE calculations were implemented and checked in Excel, especially during the initial performance evaluation phase. The ROCOF analysis also helped in understanding how dynamic signals may create challenges for PMU algorithms.

3.4 Measurement Uncertainty and Intrinsic Errors

A significant part of PMU performance evaluation is related to measurement uncertainty. A measured value is not only a number; it is affected by the device, the measurement chain, the synchronization system, the signal-processing algorithm, and the test setup. For this reason, the internship activity considered uncertainty and intrinsic errors as part of the technical background.

Possible intrinsic errors include instrument transformer errors, analog front-end gain and offset errors, ADC quantization noise, timing errors, synchronization errors, spectral leakage, and windowing effects. Some of these errors are more visible under steady-state conditions, while others become more important during dynamic tests. For example, a DFT-based phasor algorithm can produce biased estimates when the signal frequency is not exactly nominal or when the signal contains amplitude modulation.

The study of these errors was useful for interpreting the results. If a PMU shows a higher TVE under amplitude modulation than under steady-state conditions, this does not necessarily mean that the device is defective. It may reflect the difficulty of estimating phasors under dynamic conditions. Similarly, differences between the commercial micro-PMU and the INRIM reference QPMU may be related to different algorithms, different window lengths, and different internal filtering strategies.

3.5 Sub-Oscillation and SSO Analysis

Sub-synchronous oscillations and related sub-oscillation phenomena are important in modern grids, especially when renewable generation, inverter-based resources, and control interactions are present. Such phenomena may be

visible in voltage magnitude, phase angle, frequency, ROCOF, current, or active power. In this work, the focus was on PMU output quantities rather than raw waveform samples.

The PMU reporting rate imposes an important limitation. For example, with a 50 fps reporting rate, the Nyquist frequency is 25 Hz. Therefore, components above this limit cannot be directly observed without aliasing. For this reason, the developed algorithm explicitly includes a Nyquist-frequency detection step.

Another important issue is the distinction between real oscillation and slow drift or noise. PMU data can contain small fluctuations even under steady-state conditions. If an algorithm is too sensitive, it may interpret noise as an oscillation. If it is too conservative, it may miss weak oscillatory components. The developed algorithm addresses this problem by combining steady-state detection, adaptive spectral thresholding, and least-squares validation.

4. Experimental Setup and Input Data

The experimental activity involved two main measurement systems: a commercial micro-PMU and an INRIM reference QPMU. The commercial device was a Powerside PQube micro-PMU configured in M-class mode at 50 fps. The INRIM QPMU was based on a PXI measurement platform with high-resolution digitizers and a homemade synchrophasor estimation algorithm. Both systems were synchronized through GPS or GNSS-based timing.

The laboratory setup included a Tektronix arbitrary waveform generator, an amplification chain, a voltage elevator transformer, the INRIM wideband amplifier, and reference instruments for amplitude and phase checks. The waveform generator was used to produce controlled test signals. The generated signals were then acquired by the PMU systems and exported for analysis.



Figure 1. Laboratory photograph showing the waveform generator, amplifier chain, PMU/QPMU devices, and acquisition computer.

- steady-state nominal condition at 50 Hz;
- off-nominal condition at 50.05 Hz;
- amplitude modulation with 10 Hz modulation frequency and 10 percent modulation depth;

- off-nominal plus amplitude modulation;
- harmonic distortion with third and fifth harmonic components;
- PMU data containing injected or observed oscillatory components.

Excel files were used to organise and elaborate the results. MATLAB scripts were used to generate waveforms, process PMU measurements, extract dominant frequencies, and compare algorithms. The main Excel files included steady-state, off-nominal, amplitude modulation, off-nominal plus amplitude modulation, harmonic, algorithm comparison, PMU-INRIM comparison, and reporting-rate comparison files.

The MATLAB scripts were developed in two main versions. The first version was designed for a fixed PMU file format, where the relevant columns were selected by their positions. The second version was adapted for INRIM QPMU files and used preserved column names. This made the script more robust, because file exports may change the column order or delimiter.

The data preparation stage was important because incorrect column selection would lead to incorrect algorithmic results. For this reason, the scripts included checks on the number of columns, conversion of numeric values imported as text, and automatic delimiter detection in the QPMU version.

4.1 SSO Input Waveform Generation

The input waveform used for the SSO detection tests was generated in MATLAB with the script *generator_work.m*. This script creates separate arbitrary waveform files for the Tektronix AFG31000 series. The waveform follows the standard SSO model used during the internship:

$$y(t) = y_0(t) + y_1(t)$$

$$y_0(t) = A_0 \cos(2\pi f_0 t), \quad y_1(t) = A_1 \cos(2\pi f_1 t + \phi_1)$$

In this model, $y_0(t)$ is the 50 Hz fundamental component and $y_1(t)$ is the additional oscillatory component used to emulate the sub-oscillation. In the implemented test waveform, A_0 was set to 1.0, f_0 was set to 50 Hz, ϕ_1 was set to 0 rad, and A_1 was set to 5 percent of the fundamental amplitude. The

waveform was normalized to the interval $[-1, +1]$ before export so that it could be imported easily into the arbitrary waveform generator.

The sampling frequency used for waveform generation was 1 MS/s and the record length was 1 second. This choice made the generated files coherent and glitch-free for the selected frequency points, because both the 50 Hz fundamental and the tested SSO frequencies complete an integer number of cycles in the 1-second record. The tested SSO frequency points were 0 Hz, 5 Hz to 40 Hz in 5 Hz steps, and additional points around the fundamental frequency: 42, 45, 47, 48, 49, 51, 52, 53, 55, 58, 60, and 65 Hz.

For the 0 Hz case, the script sets the oscillatory term so that it becomes zero, meaning that only the fundamental component remains. For every other frequency point, the final waveform is the sum of the 50 Hz fundamental and the selected SSO component. One single-column text file is produced for each frequency, using a decimal point and CR-LF line endings, as required for reliable AFG import.

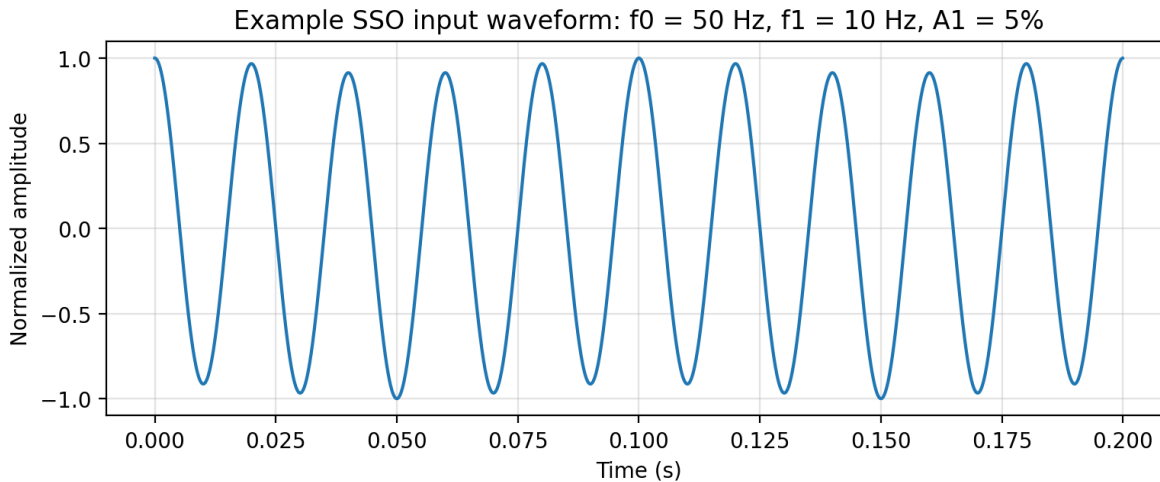


Figure 2. Example of the generated SSO input waveform used before PMU acquisition and detection. The shown example uses $f_0 = 50$ Hz, $f_1 = 10$ Hz, and $A_1 = 5$ percent of the fundamental amplitude.

This generated waveform is important for the interpretation of the detection results. The MATLAB detection algorithm does not analyse the generator file directly; instead, the waveform is first generated and applied to the measurement setup, then the PMU or QPMU output channels are acquired and processed. Therefore, the input model defines the expected oscillatory

component, while the detection algorithm estimates how that component appears after the measurement chain and PMU reporting process.

5. Literature Review and Methodological Choices

Before implementing the final analysis workflow, several scientific and technical sources were studied. The literature review focused on measurement-based oscillation analysis methods, because the available data consisted of PMU and QPMU measurements rather than a complete power-system model.

Model-based methods can be powerful when the full system model is available. However, they require accurate knowledge of the network, generators, controllers, and operating conditions. In practical monitoring applications, this information may not be available or may change over time. Measurement-based methods are therefore attractive because they use signals measured directly by PMUs. This is consistent with the objective of the internship, which was to analyse available PMU output data.

The reviewed approaches included Fourier and spectral analysis methods, Continuous Wavelet Transform methods, Prony-based methods, Matrix Pencil and modal estimation methods, EPLL-based approaches, threshold-based time-series detection methods, time-series decomposition methods, and machine-learning approaches for event classification.

Fourier-based methods are intuitive and computationally efficient. They are useful for identifying dominant frequency components in a windowed signal. Their main limitation is the trade-off between time and frequency resolution. A longer window improves frequency resolution but reduces the ability to localize events in time. A shorter window provides better time localization but may produce unstable frequency estimates.

Wavelet-based methods are more flexible for time-frequency analysis. They can track how oscillatory components evolve over time and are suitable for non-stationary signals. However, they require careful selection of wavelet parameters and may be more complex to implement and tune. In the

internship, wavelet methods were studied as a possible reference, but the implemented workflow remained based on a simpler and more transparent spectral approach.

Prony and Matrix Pencil methods are widely used for modal analysis. They can estimate frequencies and damping factors of oscillatory components. These methods are valuable when the signal can be represented as a sum of damped sinusoids. However, they may be sensitive to noise, model order selection, and preprocessing choices. For this reason, they were considered in the literature review but not selected as the main implementation for the internship workflow.

EPLL-based approaches are adaptive methods that can track oscillation parameters online. They can be useful when the objective is recursive estimation of frequency and damping. However, they require parameter tuning and a more complex implementation. The internship algorithm was instead designed to be simple, reproducible, and directly applicable to the available datasets.

The final methodological choice was therefore a sliding-window spectral algorithm with additional validation steps. This choice was motivated by the need to process several datasets, compare results, handle special cases such as 0 Hz and Nyquist frequency, and produce interpretable outputs for reporting.

6. Developed MATLAB Algorithms

6.1 General Description

Two MATLAB scripts were developed and documented: `INRIM_Algorithm_QPMU_50fps.m`, designed for PMU data read using fixed column positions, and `INRIM_algo.m`, designed for INRIM QPMU data read using preserved column names and automatic delimiter detection.

Both scripts analyse PMU data sampled at 50 frames per second. The user can select one of three channels: frequency, amplitude, or phase. The selected channel is processed using a sliding-window method. The main purpose is to estimate the dominant oscillation frequency over time. For amplitude and

phase channels, the algorithm also estimates an SSO amplitude from the original signal.

The SSO test data analysed by these scripts originated from the input waveform generated with the standard model $y(t) = y_0(t) + y_1(t)$. The known injected component f_1 was used as the reference frequency when checking whether the developed detection algorithm could correctly recover the dominant oscillatory component from PMU output channels.

Parameter	Value	Description
Fs	50 fps	PMU or QPMU reporting rate
windowSec	6 s	Length of the analysis window
stepSec	1 s	Step between consecutive windows
samples per window	300	Number of samples analysed in each window

A new frequency estimate is produced approximately every second. This means that the output is a time sequence of detected dominant frequencies. The timestamp associated with each estimate corresponds to the centre of the analysis window.

6.2 Input Handling and Data Robustness

The fixed-column version reads the input file using `readcell` and `readmatrix`. It expects the timestamp in the first column, the phase in the third column, the amplitude in the fourth column, and the frequency in the fifteenth column. This version is suitable when the CSV export has a stable structure.

The INRIM QPMU version reads the file using preserved column names. This is more robust because the algorithm searches for the required variables by their names rather than by fixed positions. The required columns are timestamp, INRIM QPMU voltage angle, INRIM QPMU voltage magnitude, and INRIM QPMU frequency. The script also detects whether the file uses commas, semicolons, or tabs as delimiters.

This part of the work was important because data analysis errors can originate from simple file-reading problems. For example, a numeric column can be

imported as text, or a file can be exported using a delimiter different from the expected one. The helper functions for delimiter detection and numeric conversion made the script more reliable.

6.3 Algorithm Steps

- remove the mean or median component from the signal;
- check whether the window is nearly constant and should be classified as 0 Hz;
- check whether the signal strongly matches the alternating pattern corresponding to the Nyquist frequency;
- detrend the signal and apply a Hann window;
- compute the FFT and search for dominant spectral peaks between approximately 0.2 Hz and 24.8 Hz;
- apply adaptive thresholds based on the spectrum median and MAD;
- select the strongest valid peak and validate it using a sinusoidal least-squares fit;
- store the detected frequency and, for amplitude or phase channels, the SSO amplitude at the centre timestamp of the window.

For amplitude and phase channels, the detection step uses a normalized signal. However, the reported SSO amplitude is calculated from the original non-normalized signal in order to preserve physical meaning. This is an important design choice because normalization improves numerical robustness, but it would make the amplitude output less meaningful if it were also used for amplitude reporting.

6.4 Special Cases and Validation

The first special case handled by the algorithm is the steady-state condition. If the signal variation inside the window is very small, the detected frequency is assigned as 0 Hz. This avoids interpreting measurement noise or very small fluctuations as a meaningful oscillation.

The second special case is the Nyquist-frequency condition. Since the sampling rate is 50 fps, the Nyquist frequency is 25 Hz. A component at this

frequency appears as an alternating pattern. The algorithm checks this behaviour and assigns 25 Hz when the Nyquist pattern is dominant.

If the window is neither steady-state nor Nyquist-dominated, the algorithm performs ordinary spectral detection. The signal is detrended, multiplied by a Hann window, and transformed using the FFT. The Hann window reduces spectral leakage, which is important when the signal frequency does not fit an integer number of cycles inside the analysis window.

After a candidate frequency is selected from the spectrum, the algorithm validates it using a sinusoidal least-squares fit. The idea is to check whether a sinusoid at the candidate frequency can explain a meaningful portion of the signal in that window. This validation step reduces false positives caused by isolated spectral artefacts or noise.

Before processing the phase channel, the algorithm unwraps the phase to avoid artificial discontinuities due to angle wrapping. Phase signals can jump from positive to negative values when crossing the wrapping boundary. These artificial jumps create false high-frequency components in the spectrum. Unwrapping removes this problem and makes the phase signal smoother and more suitable for frequency-domain analysis.

Selected channel	Output file	Main quantities
Frequency	frequency_channel_results.csv	Timestamp and detected frequency
Amplitude	amplitude_channel_results.csv	Timestamp, detected frequency, SSO amplitude
Phase	phase_channel_results.csv	Timestamp, detected frequency, SSO amplitude

7. Results and Discussion

7.1 Steady-State Tests

The steady-state tests were used to verify the basic behaviour of the PMU systems under nominal 50 Hz conditions. This condition is important because

it represents the reference case. Under ideal steady-state conditions, the PMU should report stable magnitude, phase, frequency, and low TVE values.

The commercial micro-PMU was configured in high-precision M-class mode at 50 fps. The INRIM reference QPMU was evaluated using different sliding-overlapped-window configurations. The comparison showed that the selected window length can influence the measurement result. In general, longer windows provide more stable phasor estimates because they include more samples, but they also introduce more latency and may reduce responsiveness to fast changes.

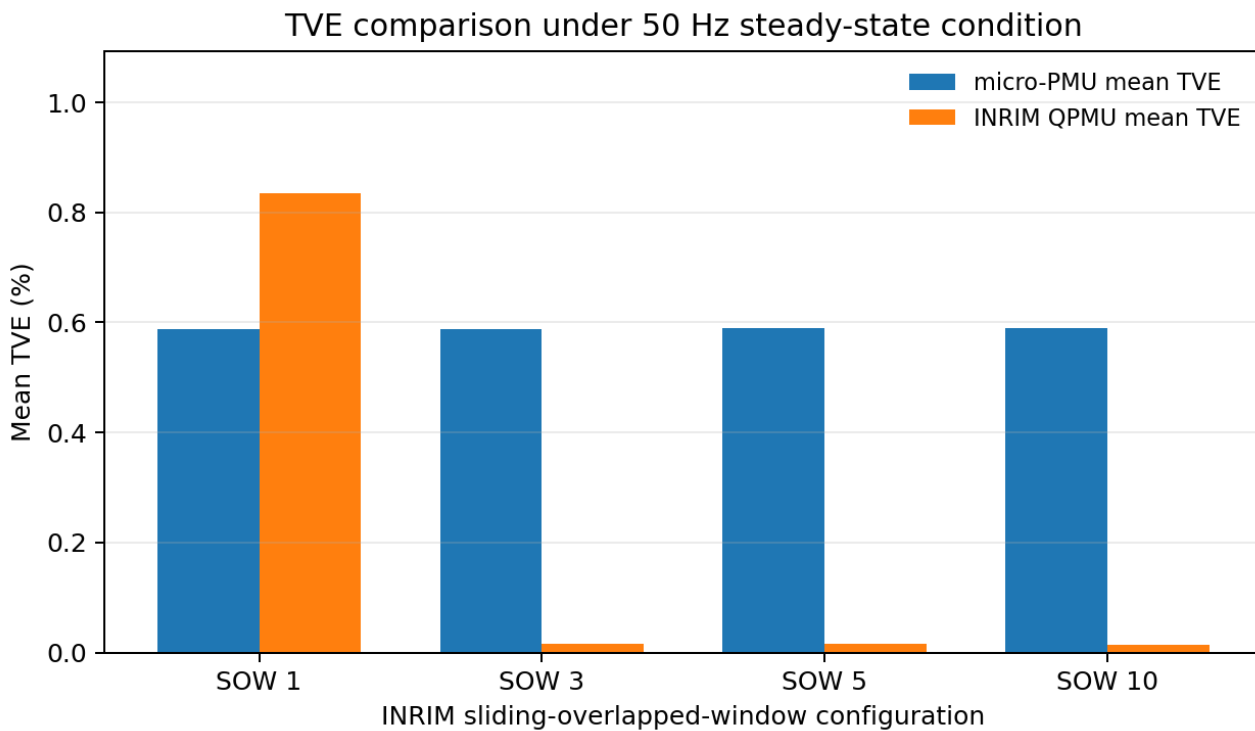


Figure 2. Mean TVE comparison under 50 Hz steady-state conditions. The chart is generated from *steady_state_elaboration.xlsx* and compares the commercial micro-PMU with the INRIM QPMU configurations.

The interpretation of this graph focuses on whether the systems remain within the expected performance limits and how the reference QPMU changes with the selected sliding-overlapped window. The figure also supports the discussion that PMU performance evaluation depends not only on the device but also on the signal-processing settings.

7.2 Off-Nominal Tests

The off-nominal tests used a fundamental frequency of 50.05 Hz. This condition is important because practical power systems do not always operate exactly at nominal frequency. Even small deviations from nominal frequency can affect phasor estimation algorithms, especially if they are based on fixed nominal-frequency assumptions.

The off-nominal condition can create spectral leakage when the analysis window does not contain an integer number of cycles. It can also affect TVE and frequency estimation. For this reason, the off-nominal test was used to evaluate how the commercial micro-PMU and the INRIM QPMU behave when the fundamental frequency is slightly shifted.

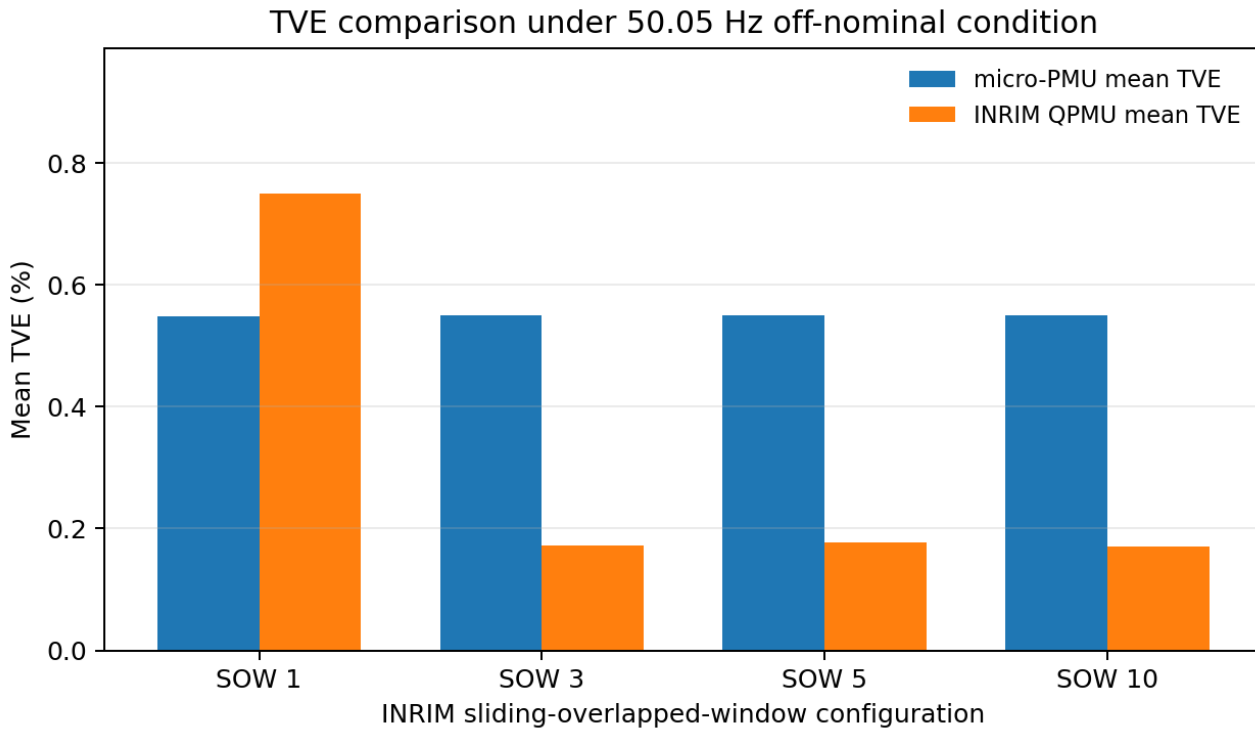


Figure 3. Mean TVE comparison under 50.05 Hz off-nominal conditions. The chart is generated from *off-nominal_elaboration.xlsx* and shows the influence of a small deviation from nominal frequency.

The off-nominal results should be interpreted together with the steady-state case. If TVE increases, the increase can be related to algorithmic sensitivity to off-nominal frequency, windowing effects, and spectral leakage. If the values remain low, this indicates good robustness under small frequency deviations.

7.3 Amplitude Modulation Tests

Amplitude modulation tests were performed using a 50 Hz fundamental component, a 10 Hz modulation frequency, and 10 percent modulation depth. These tests were important because modulation changes the amplitude of the measured signal over time and may affect phasor estimation.

In PMU output data, amplitude modulation may appear clearly in the voltage magnitude channel. The developed algorithm was therefore applied to the amplitude channel to estimate the dominant oscillation frequency and the associated SSO amplitude. The expected detected oscillation frequency for this test is related to the modulation frequency, although the exact observed behaviour can depend on the PMU algorithm and reporting rate.

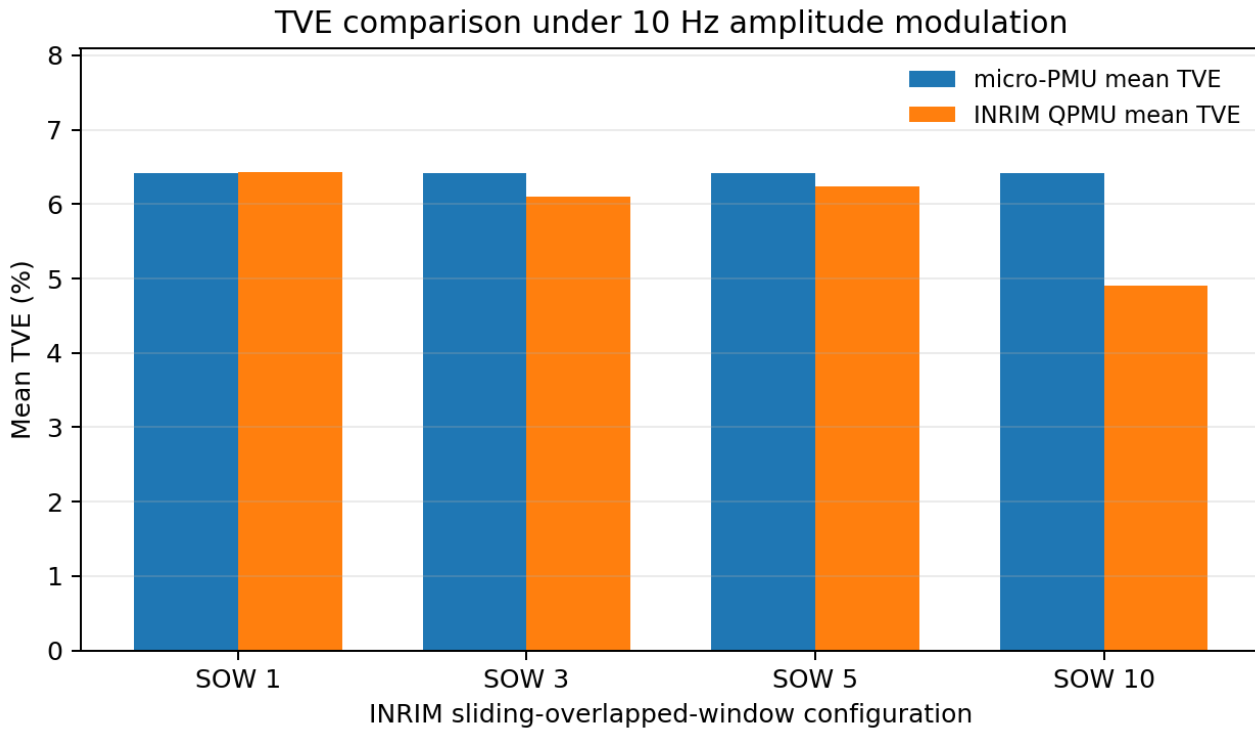


Figure 4. Mean TVE comparison under the 10 Hz amplitude modulation condition. The chart is generated from *Amplitude_Modulation_elaboration.xlsx* and summarizes the PMU and QPMU response under a dynamic amplitude variation.

The figure shows that amplitude modulation creates a more challenging dynamic condition than steady-state operation. It also supports the use of the developed algorithm for recovering oscillatory information from the amplitude channel.

7.4 Off-Nominal Plus Amplitude Modulation

The combined off-nominal plus amplitude modulation test used a 50.05 Hz fundamental frequency with 10 Hz modulation and 10 percent modulation depth. This condition is more challenging because it combines two effects: a deviation from nominal frequency and a dynamic amplitude variation.

This test is useful for evaluating the robustness of both the PMU measurement system and the developed detection algorithm. The PMU must estimate phasors when the fundamental frequency is not exactly nominal, and the algorithm must identify the oscillatory component from the output channel without confusing it with the off-nominal behaviour.

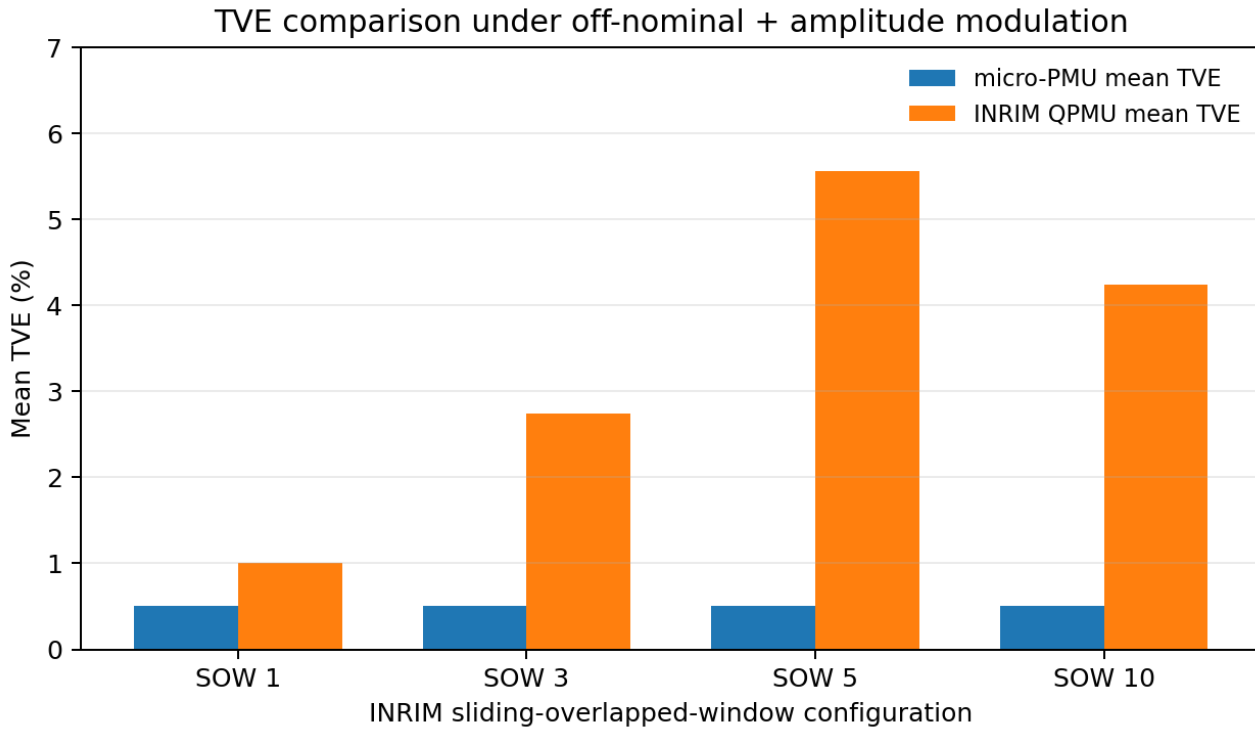


Figure 5. Mean TVE comparison under combined off-nominal plus amplitude modulation conditions. The chart is generated from *off_nominal+AM_elaboration.xlsx* and highlights the combined effect of frequency deviation and amplitude modulation.

This test represents a more realistic dynamic scenario than the single-effect tests. A robust algorithm must avoid confusing fundamental-frequency deviation with modulation-related oscillation, while still detecting the dominant oscillatory component in the selected PMU channel.

7.5 Harmonic Distortion Tests

Harmonic distortion tests used a 50 Hz fundamental signal with third and fifth harmonic components. Harmonics may affect PMU phasor estimation and may create disturbances in PMU output quantities. These tests were therefore useful for checking whether the developed algorithm avoided false detections produced by harmonic-related measurement artefacts.

Although the injected harmonics are components of the raw waveform, the developed algorithm works mainly on PMU output quantities. Therefore, the effect of harmonics is observed indirectly through their influence on estimated phasors, frequency, and other reported channels. This distinction is important because the algorithm is not a raw waveform harmonic analyser; it is a PMU-output analysis tool.

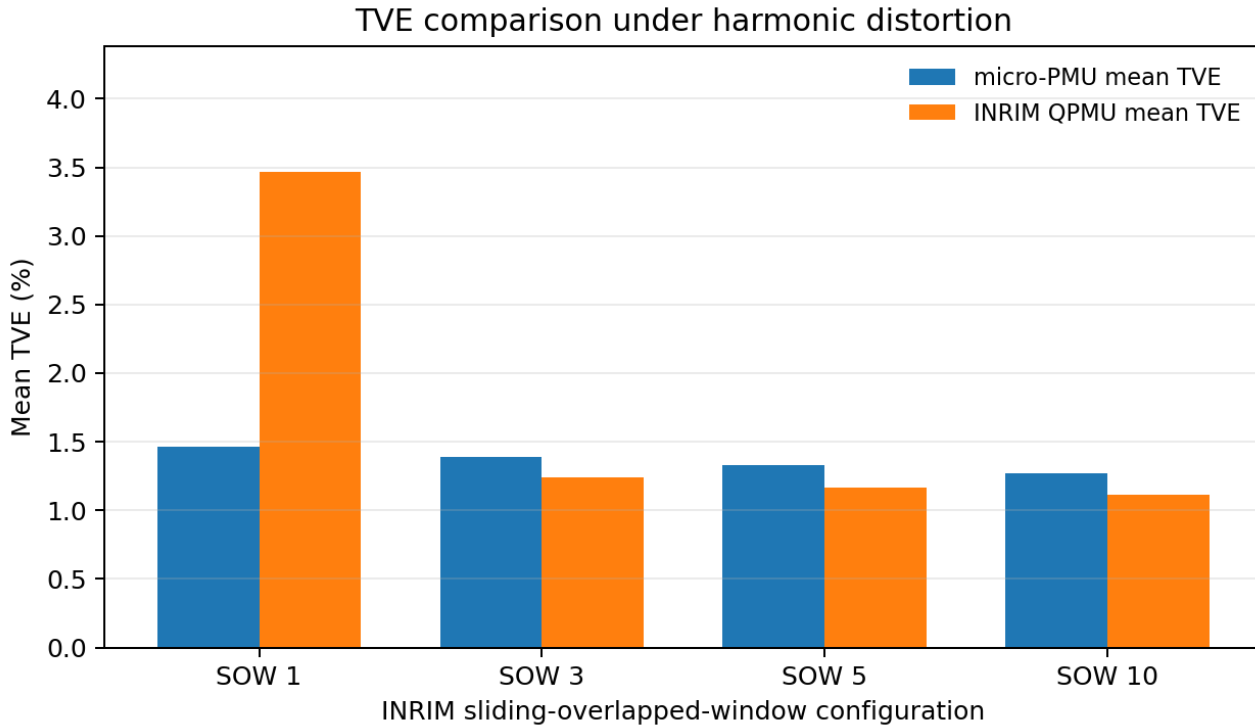


Figure 6. Mean TVE comparison under harmonic distortion. The chart is generated from *harmonics_elaboration.xlsx* and summarizes the effect of third and fifth harmonic components on PMU/QPMU measurements.

The harmonic distortion results help evaluate whether harmonics increase TVE and whether they create patterns in the PMU outputs that could be

misinterpreted as oscillatory behaviour. This is important for the reliability of automated detection algorithms.

7.6 NPL and INRIM Algorithm Comparison

The developed INRIM algorithm was compared with the NPL algorithm. This comparison was used to check consistency between the developed method and an external reference approach. The comparison included PMU and INRIM QPMU measurements and different reporting-rate conditions.

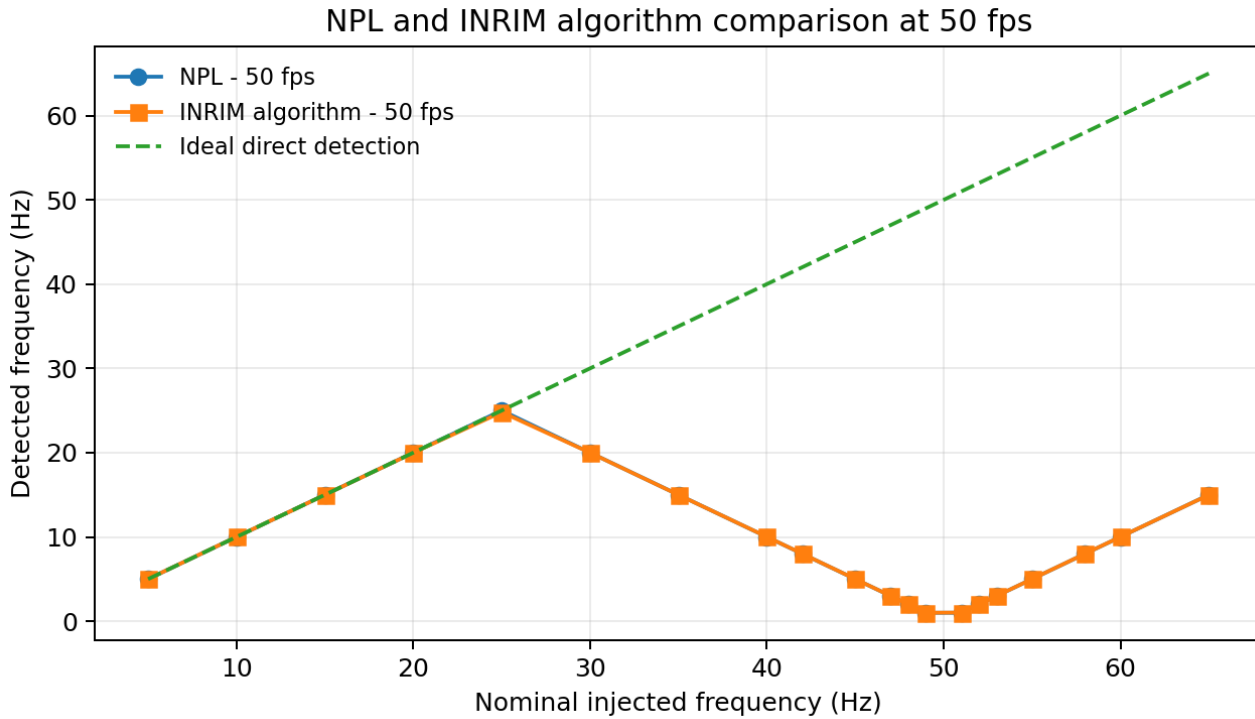


Figure 7. Detected-frequency comparison between NPL and INRIM algorithms at 50 fps. The chart is generated from `algorithm_comparison_NPL_INRIM.xlsx`.

Agreement between the two algorithms supports the validity of the developed workflow. Differences should be analysed carefully, because they may reveal different thresholding choices, different windowing strategies, different treatment of edge cases, or different sensitivity to noise. Frequencies above the Nyquist frequency appear through their observable aliased components when the reporting rate is not sufficient.

7.7 Reporting-Rate Comparison

The effect of reporting rate was analysed using micro-PMU measurements at different frame-per-second values. This aspect is important because the maximum observable frequency depends on the reporting rate. Lower reporting rates reduce the range of observable oscillatory components and increase aliasing risk.

Reporting rate	Nyquist frequency
10 fps	5 Hz
25 fps	12.5 Hz
50 fps	25 Hz
100 fps	50 Hz

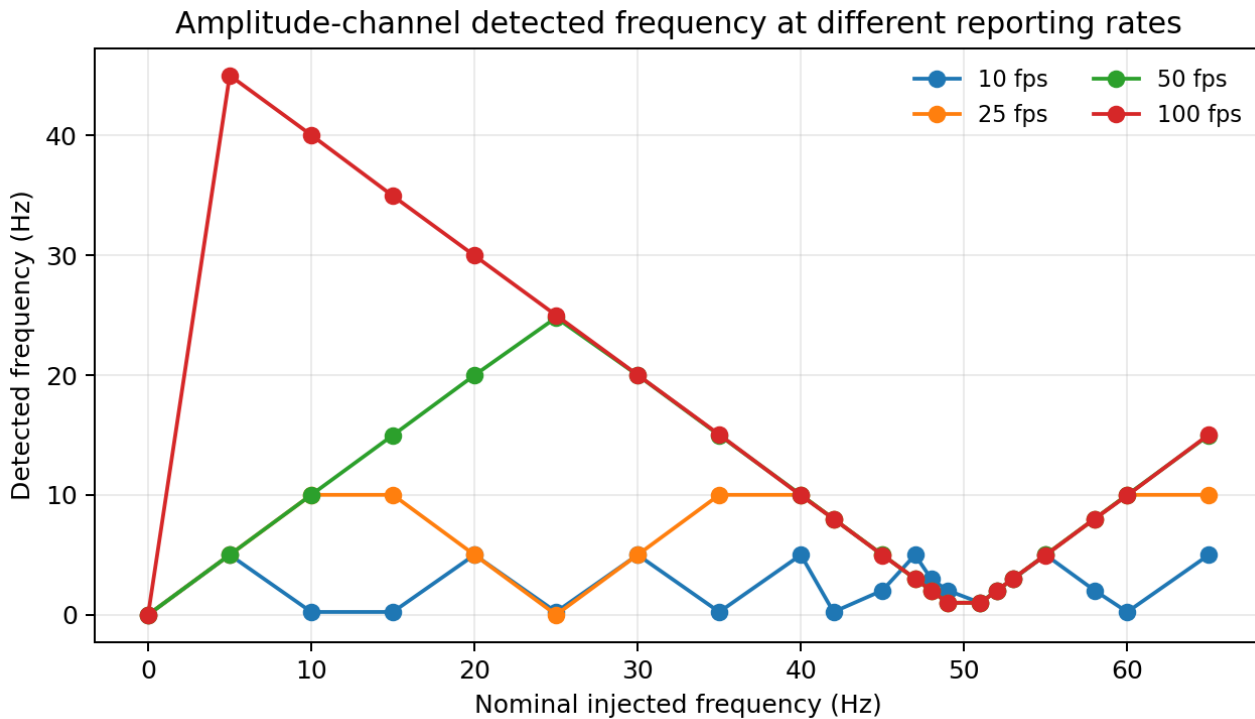


Figure 8. Amplitude-channel detected-frequency comparison at different reporting rates. The chart is generated from *Amplitude_channel_comparison_with_different_fps_microPMU.xlsx*.

The chart shows why reporting rate must be considered when interpreting detected frequencies. A 10 fps signal cannot directly represent a 10 Hz oscillation because its Nyquist frequency is only 5 Hz. A 50 fps signal can

represent frequencies up to 25 Hz, while a 100 fps signal can represent frequencies up to 50 Hz.

7.8 PMU and INRIM QPMU Comparison

The developed algorithm was applied to both commercial PMU data and INRIM QPMU data. This comparison was important because it allowed the same detection method to be applied to two measurement systems. The commercial micro-PMU and the INRIM reference QPMU may differ in hardware, internal algorithms, filtering, and data export format.

The INRIM QPMU version of the script was made more robust by reading data using preserved column names and automatic delimiter detection instead of relying only on fixed column positions. This improvement reduced the risk of selecting wrong columns and made the algorithm more suitable for repeated analysis of exported data.

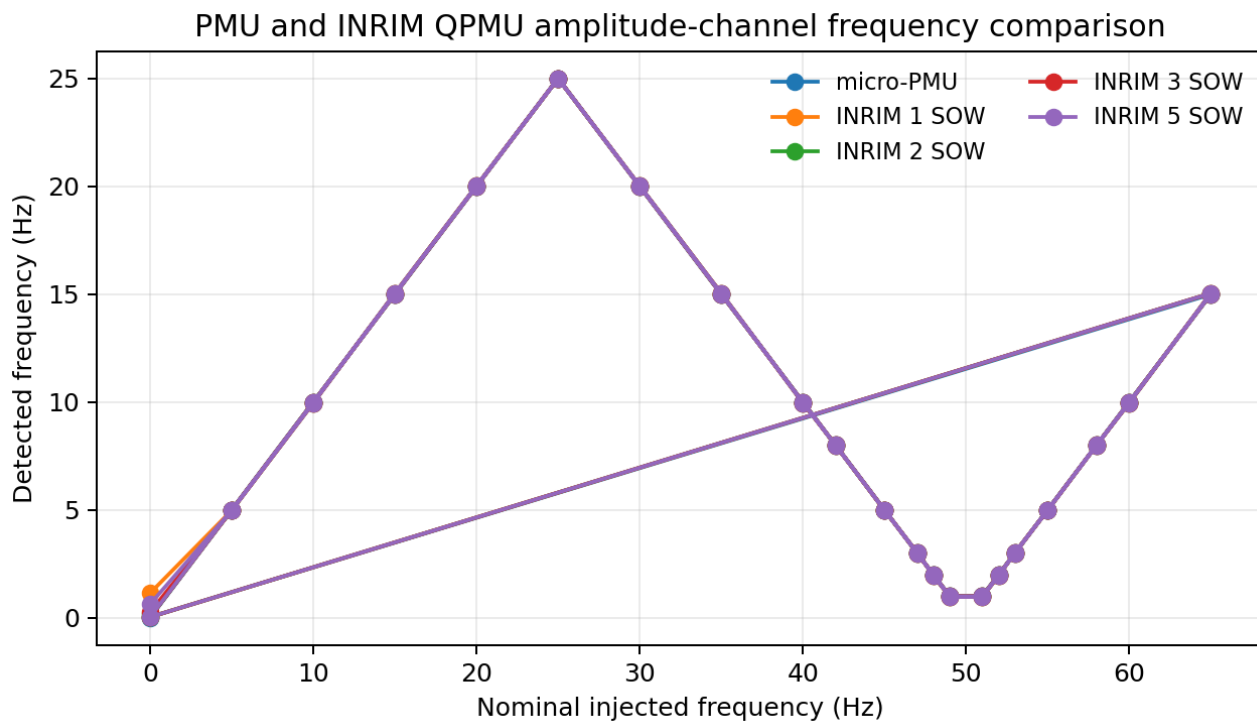


Figure 9. PMU and INRIM QPMU amplitude-channel frequency comparison. The chart is generated from `comparison_PMU_INRIM_SOW_NPL_INRIM_ALGO.xlsx`.

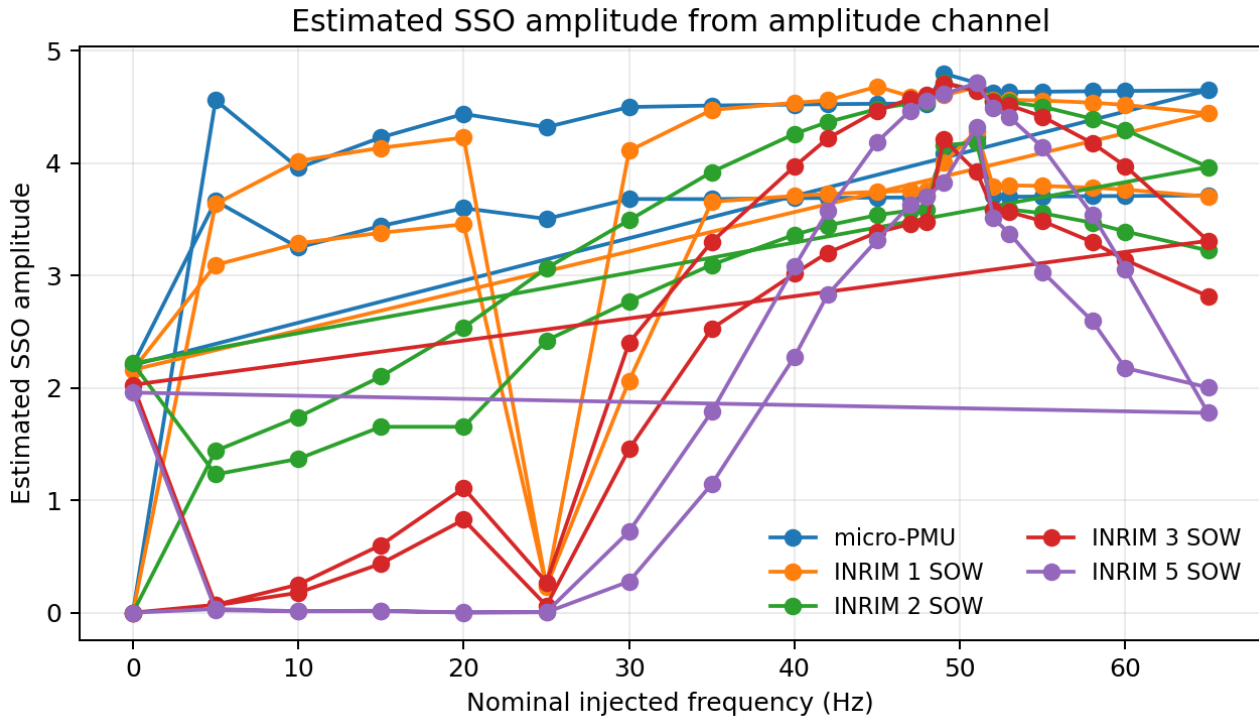


Figure 10. Estimated SSO amplitude comparison for micro-PMU and INRIM QPMU configurations. The chart is generated from `comparison_PMU_INRIM_SOW_NPL_INRIM_ALGO.xlsx`.

Differences between PMU and QPMU results may be related to device behaviour, reporting rate, processing window, and reference algorithm settings. The comparison is not only a numerical exercise; it helps identify which configurations provide more stable and reliable oscillation estimates.

8. Main Contribution of the Internship

The main technical contribution of this internship was the development of a MATLAB-based workflow for PMU and QPMU data analysis. The algorithm was not copied directly from a single paper. Instead, it was designed after studying several measurement-based oscillation analysis methods and adapting the most suitable ideas to the available data.

The final workflow combines robust file reading, channel selection, sliding-window analysis, steady-state detection, Nyquist-frequency detection, FFT-based dominant peak extraction, MAD-based adaptive thresholding, sinusoidal least-squares validation, amplitude extraction from the original

non-normalized signal, and comparison between commercial PMU, INRIM QPMU, and NPL results.

This contribution is relevant for Smart Grid Data because it provides a reproducible method for extracting oscillatory information from synchronized measurement data. It also creates a bridge between the literature review and practical data processing. The literature provided the theoretical background, while the MATLAB scripts translated those concepts into an operational analysis workflow.

Another important contribution was the adaptation of the algorithm to different data formats. In real measurement activities, data files are not always perfectly standardized. Columns may have different names, delimiters may change, and numeric data may be imported as strings. The QPMU version of the script addressed these issues by using preserved variable names, automatic delimiter detection, and numeric conversion.

The internship also contributed to the comparison between different measurement systems and algorithmic approaches. The PMU, QPMU, NPL algorithm, and INRIM algorithm were considered together in the final analysis. This comparison is valuable because it helps identify not only whether an oscillation is detected, but also how detection depends on the measurement system and processing method.

9. Limitations and Future Work

The developed method is based on windowed analysis. Therefore, the selected window length creates a trade-off between frequency resolution and time localization. A 6-second window provides stable spectral estimates, but very short events may be smoothed or partially missed. A shorter window could improve time localization but would reduce frequency resolution and increase uncertainty in the detected peak.

The algorithm mainly reports the dominant frequency in each window. If several components exist simultaneously, weaker components may not be selected. Future work could include multi-peak extraction and tracking. This

would make the algorithm more suitable for signals containing multiple oscillatory modes.

Another important limitation is the reporting rate. Since the method uses PMU output quantities, the observable frequency range is limited by the PMU reporting rate. For 50 fps data, the maximum direct frequency is 25 Hz. If the physical phenomenon contains components above this limit, those components may be aliased or not visible in the PMU output.

The algorithm also depends on the selected channel. An oscillation that is clearly visible in the amplitude channel may be weak in the frequency channel, or vice versa. Future versions could combine multiple channels to improve detection robustness. A multi-channel approach could compare frequency, amplitude, phase, and ROCOF simultaneously and assign a confidence level to each detected event.

Future work could also include damping estimation. Some literature methods, such as Prony, Matrix Pencil, and EPLL-based approaches, can estimate damping in addition to frequency. Adding damping estimation would make the algorithm more informative for power-system stability analysis. However, it would also require additional validation and careful handling of noisy PMU data.

Finally, the report-generation workflow could be further automated. Excel tables, MATLAB outputs, and final graphs could be connected in a single reproducible pipeline. This would reduce manual effort and improve consistency between calculations, figures, and written documentation.

10. Conclusions

The internship at INRiM provided practical experience in PMU measurement evaluation, Smart Grid Data analysis, and MATLAB-based algorithm development. The work began with the study of synchrophasor measurement principles, TVE, ROCOF, measurement errors, and dynamic PMU test conditions. It then developed into an algorithmic activity focused on sub-oscillation detection and dominant frequency extraction.

A MATLAB workflow was implemented to process frequency, amplitude, and phase channels. The algorithm uses sliding windows, special handling for steady-state and Nyquist-frequency cases, FFT-based peak detection, adaptive thresholds, and least-squares validation. It was applied to both commercial PMU data and INRIM QPMU data, and it was compared with the NPL algorithm and with results obtained at different reporting rates.

The work showed that PMU-based oscillation analysis requires careful attention to reporting rate, channel selection, window length, thresholding, and measurement uncertainty. It also showed that an algorithm developed from literature concepts must be adapted to the practical constraints of real data files and measurement systems.

Overall, the internship contributed to a structured and reproducible analysis workflow for synchronized PMU data. The developed algorithms, Excel elaborations, and comparison figures provide a useful basis for future work on Smart Grid Data analysis and automated sub-oscillation detection.

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[8] INRiM internal measurement files and MATLAB scripts used during the Smart Grid Data internship activity.

Appendix A - Figure and Data Sources

Figure	Source file	Content
Figure 2	steady_state_elaboration.xlsx	TVE comparison under 50 Hz steady-state condition
Figure 3	off-nominal_elaboration.xlsx	TVE comparison under 50.05 Hz off-nominal condition
Figure 4	Amplitude_Modulation_elaboration.xlsx	TVE comparison under 10 Hz amplitude modulation
Figure 5	off_nominal+AM_elaboration.xlsx	TVE comparison under off-nominal plus AM condition
Figure 6	harmonics_elaboration.xlsx	TVE behaviour under harmonic distortion
Figure 7	algorithm_comparison_NPL_INRIM.xlsx	NPL algorithm vs developed INRIM algorithm
Figure 8	Amplitude_channel_comparison_with_different_fps_microPMU.xlsx	Amplitude-channel comparison at different reporting rates
Figures 9-10	comparison_PMU_INRIM_SOW_NPL_INRIM_ALGO.xlsx	PMU vs INRIM QPMU comparison and SSO amplitude comparison

Appendix B - Algorithm Pseudocode

Input: PMU or QPMU data file, selected channel, $F_s = 50$ fps

Output: timestamped detected frequency and SSO amplitude when applicable

1. Read the input file.
2. Extract timestamp, frequency, amplitude, and phase channels.
3. Convert numeric columns if needed.
4. Ask the user to select the analysis channel.

5. If the phase channel is selected, unwrap the phase.
6. Divide the signal into 6-second sliding windows with 1-second step.
7. For each window:
 - a. Remove mean or median component.
 - b. Check steady-state condition.
 - c. Check Nyquist-frequency condition.
 - d. Detrend the signal.
 - e. Apply Hann window.
 - f. Compute FFT.
 - g. Search for peaks in the valid frequency range.
 - h. Apply median and MAD thresholding.
 - i. Select the strongest valid peak.
 - j. Validate the peak using sinusoidal least-squares fitting.
 - k. Compute SSO amplitude from the original signal.
 - l. Store the result at the centre timestamp.
8. Export the result table.
9. Generate plots.